Isaac Mubiru

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Kampala, Uganda · EAT (UTC+3) · Remote, US East Coast overlap

PROFESSIONAL SUMMARY

Backend engineer (6+ years in production) who builds **data pipelines and third-party integrations** on **Google Cloud Platform**. Kept **real-time stock market prices** current from Morningstar APIs across 2M+ records for a retail-investing app; built **real-time two-way sync between two live datastores on GCP Pub/Sub**; led a **zero-downtime migration of 20+ production apps** reporting directly to the CTO. 4+ years on remote teams for US companies (MTailor, Mind2matter) from East Africa. Now looking to apply the same integration and reliability work to trading infrastructure: new exchanges and data sources, normalised feeds and live monitoring. Code: github.com/zac-09

CORE COMPETENCIES

Market Data & Third-Party API Integration | Real-Time Data Pipelines (Pub/Sub, Kafka) | Google Cloud Platform | Backend Systems in Node.js / TypeScript | Batch Jobs over Large Datasets | Zero-Downtime Migrations | Microservices: Kafka, gRPC, Docker, Kubernetes | Small-Team Ownership, CTO-Facing

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer (US Remote, reports to CTO)

- Implemented **real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub** for message processing, keeping two live production datastores consistent during the migration.
- Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Saved the company **$5,000/month** by migrating all services off AWS onto GCP.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a Python script.
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion; implemented Express Shipping, adding **$40 revenue per order**.

Stack: Node.js, GCP (Pub/Sub, Cloud Storage), Firebase/Firestore, MongoDB, Python, AWS, ffmpeg, React

MTailor (Contractor)

Mar 2022 – Jul 2022

Backend Migration Contractor (US Remote)

- Built a migration script reading data from MongoDB and writing to Firestore after running complex processing logic.
- Wrote documentation for fellow engineers on the new Firebase SDKs (Node.js and Python); trained the Ops team on the Firebase dashboard and backup procedures.

Stack: Node.js, Python, MongoDB, Firebase, Google Cloud Platform, AWS

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Led optimisation of a PWA delivering **real-time stock market prices** from a Firebase backend.
- Built ad-hoc jobs querying **over 2 million records** to keep customer prices current from **Morningstar market-data APIs**.
- Extended backend APIs in **TypeScript** using Firebase Cloud Functions and Express.js.

Stack: Node.js, TypeScript, Firebase, Express.js, React

Mind2matter

Jan 2022 – Jul 2022

Software Engineer (US Remote, reports to CTO)

- Built backends for DeFi applications using Web3 and Node.js for a US agency with demanding clients.
- Delivered React UIs from Figma designs under tight timelines.

Stack: Node.js, Web3, Firebase, React

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead (Uganda Remote)

- Architected a **microservices backend using Apache Kafka, Docker and Kubernetes** for FIDA Uganda's case management app.
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC.
- Led a team of 4 developers building legal-tech systems for NGOs; shipped a React Native + Expo mobile app (iOS & Android).

Stack: Node.js, Apache Kafka, NATS Streaming, gRPC, Docker, Kubernetes, React Native, Firebase

PROJECTS

Parse → Firebase/GCP Platform Migration (MTailor) 20+ apps, zero downtime

Owned end to end, from plan through cutover: live MongoDB↔Firestore sync on Pub/Sub so both systems served production in parallel, per-app migration, SDK docs and Ops training. Closed with a $5,000/month saving from leaving AWS.

Node.js, Python, GCP Pub/Sub, MongoDB, Firestore, Cloud Storage

Career-ops — Automated Scanning & Verification Pipeline scheduled automation + monitoring

Node.js automation that scans dozens of external sources on a schedule, normalises results into one format, deduplicates against history, verifies each item live before acting, and logs every rejection with a reason so yield per source can be tracked and noisy sources disabled.

Node.js, Playwright, YAML, Claude Code, Git

Event-Driven Microservices Backend (FIDA Uganda) Kafka + Kubernetes

Service boundaries around Kafka events, containerised with Docker and deployed on Kubernetes, gRPC for synchronous calls; served web, mobile and USSD clients.

Node.js, Apache Kafka, gRPC, Docker, Kubernetes

EDUCATION

B.Sc. Computer Engineering — Makerere University — Aug 2018 – Sep 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript, JavaScript, GCP, Firebase, Python, MongoDB, SQL, Git, Linux, React
Data & infra: Google Pub/Sub, Cloud Storage, Apache Kafka, NATS Streaming, gRPC, Docker, Kubernetes, Express.js, AWS
Intermediate: Go, Rust, React Native
Languages: English (fluent, 4+ yrs on remote US teams), Kiswahili